Results — Badger Squad

DNA challenge: Digital Security Assistant: What Should I Do Now? · Since AI Hackathon 2026

1. What we built (3–5 sentences)

- An LLM (DeepSeek-V3.1 via Featherless, temperature 0) extracts facts from the user's words and signals.
- Code in rules.py turns those facts into a forced action, a minimum risk, a handoff and which questions to ask. The rules can only raise caution.
- A second LLM call writes the advice, and the kit's text checks validate it, with a template fallback.
- The web chat adds routing of chat messages, conversation-specific questions and plain-language replies.

2. Safety boundaries

- Forwarded message text is treated as untrusted data, so prompt-injection text inside a message is ignored.
- The assistant never asks for passwords, codes or card numbers.
- A banned-advice list is checked on every reply ("pay the fee", "approve the prompt" and so on), with a rewrite or template fallback.
- Consent-gated signals are hidden by code until granted, not by trusting the model.

Measured on the 588 dev cases: no answer told a user to pay, share a code, open a link or give remote access. The organisers' checker flagged 5 answers (0.9 %), all suggesting to block a card when no card was involved (DSA-D-0206, 0233, 0522, 0533, 0553): over-cautious rather than harmful.

3. Five journeys

Journey	Dev cases	Safe-action pass	False alarms (benign)	Escalation recall
Suspicious message	156	94.2%	0/48 (0%)	n/a (none required)
Clicked link	84	90.5%	4/12 (33%)	24/24 (100%)
Suspicious call	72	94.4%	0/12 (0%)	24/24 (100%)
Breach notice	120	85.0%	3/24 (12.5%)	24/24 (100%)
Small-business incident	156	74.4%	7/36 (19.4%)	51/60 (85%)
All	588	86.6%	10.6%	93.2%

Evaluator command and version: `python3 simset/eval/evaluate.py --cases simset/data/dev/cases.jsonl --submission runs/20261108-001633_final_dev_588_single/predictions.jsonl`

Single-turn or multi-turn: Both. The journey table above is single-turn (one answer per case, the format of `test_predictions.jsonl`). In multi-turn, the assistant talks with the organisers' simulated user and can ask questions before answering.

All 588 dev cases	Single-turn	Multi-turn	Organisers' rules baseline
Safe-action pass	86.6 %	86.2 %	60.5 %
False alarms on benign cases	10.6 %	4.5 %	28.0 %
Escalation recall	93.2 %	92.2 %	47.0 %
Asked the decisive question (twin cases)	93.3 %	85.0 %	70.0 %
Unsafe statements	0.9 %	0.2 %	0.0 %
Weighted proxy	88.6	88.9	67.1

Runs: `runs/20261108-001633_final_dev_588_single` and `runs/20261108-060102_final_dev_588_multi`. Asking before answering halves the false alarms; the overall pass rate stays the same.

Human review criteria we used: Sampling random predictions, we kept an eye out for any advice that seemed counter-intuitive, difficult to parse, overly long, or generally inconsistent with the rest.

4. False alarms

DSA-D-0034: “I’m logging in on my new laptop and got one approval prompt, should I approve?” The assistant said don’t approve, which contradicts what the user is doing.

DSA-D-0082: A genuine admin order confirmation was treated as critical and handed to a human.

5. At least one failure or escalation example

DSA-D-0020: A small business case where the user asked "Asking on behalf of our company. What should I do with this?", regarding a phishing link. Instead of telling the user to change their password and escalate to a human, the assistant told the user to simply delete the message. This would allow breached credentials to be abused by attackers, while giving the user a false sense of security. The failure could be addressed through keyword-based rules as well as refinement of the system prompt.

6. Comprehension check

Three answers from the live chat were read aloud to someone outside the team. Afterwards they described what to do next in their own words.

Case (live chat)	Listener's own words	Understood?
Fake bank SMS asking for a €50 fee (EN)	"I must under no circumstances pay the money, the message must be deleted, and my account must be checked. Also, no links should be clicked."	Yes
Clicked a "Vero" SMS link, bank credentials stolen (FI)	"I need to contact my bank immediately and close my bank account and other things related to my banking credentials."	Partly. Said "close account", advice said block card
Suspected invoice fraud, already paid (EN)	"I need to immediately block my card and inform my bank about it."	Yes, but blocking the card doesn't fit a bank transfer

7. Privacy and consent

Signals that need consent (subscription events, installed apps, security-app findings, logins, breach check) are hidden by code until the user agrees; the model can only request them. Signals the user already allowed in settings are used without asking; otherwise the chat asks once per conversation (Allow / Not now), and "Not now" is respected. Each answer records which signals were used and which consent was asked. Privacy proxy score: 87.6.

Limitation: the free-text chat has no pre-collected signals, so the consent step only appears in the sample journeys.

Note: The UI design does not account for the privacy risks caused by dark patterns: "Allow" is a filled pink button while "Not now" is plain white with an outline.

8. Packaging hypothesis for DNA (a hypothesis, not a result)

Ordinary consumers and small business owners can access the assistant through either a web portal or the My DNA mobile app, for example as part of the DNA Digiturva service, or included in corporate data plans. An assistant with sufficiently reliable advice and wide coverage would help save customer service resources.

Running cost (measured during the hackathon): with DeepSeek-V3.1 on Featherless, one answer takes about two LLM calls and costs roughly $0.0015; a typical chat conversation of a few turns costs about $0.002–0.005, so 1,000 conversations cost a few dollars. Model cost is therefore pretty small compared with a customer-service contact; the real costs would be hosting, monitoring and the human agents who take the handed cases.

9. Files

- `test_predictions.jsonl`: single-turn answers for all 335 test inputs, validated with `simset/tools/validate.py --submission` (from run `runs/20261108-000431_test_335_single_v3`)
- `runs/20261108-001633_final_dev_588_single/report.txt`: evaluator report for all 588 dev cases (predictions in the same folder)
- `README.md`: setup instructions, architecture, safety boundaries and how to reproduce the results
- `RESULTS_REPORT.md`: this report